

# 1 Graph Theoretic Foundations & Traffic Dynamics

The urban transportation network is modeled as a weighted directed graph, serving as the foundation for the physical routing layer.

- **Graph Definition:** Let  $G = (V, E)$  be a directed graph where  $V = \{0, 1, \dots, N\}$  is the set of nodes (intersections/depots) and  $E = \{(i, j) \mid i, j \in V\}$  is the set of physical road edges. Node 0 represents the origin depot.
- **Distance Matrix:**  $D \in \mathbb{R}^{(N+1) \times (N+1)}$  where  $D_{ij}$  is the physical length of edge  $(i, j)$  in meters.
- **Free-Flow Time Matrix:**  $T_{\text{free}} \in \mathbb{R}^{(N+1) \times (N+1)}$  representing unobstructed travel times.
- **Dynamic Congestion Tensor:**  $\rho \in \mathbb{R}^{(N+1) \times (N+1)}$  where each element  $\rho_{ij} \geq 1.0$  represents a randomized dynamic traffic penalty.
- **Effective Travel Time:** The framework updates the final weight matrix  $T \in \mathbb{R}^{(N+1) \times (N+1)}$  dynamically:

$$T_{ij} = T_{\text{free},ij} \cdot \rho_{ij} \quad (1)$$

## 2 MILP Formulation of CVRPTW

The core optimization problem is the Capacitated Vehicle Routing Problem with Time Windows (CVRPTW), solved over the computed shortest-path matrix.

### 2.1 Decision Variables

- $x_{ijk} \in \{0, 1\}$ : Binary tensor; equals 1 if vehicle  $k \in K$  traverses the path from node  $i$  to  $j$ , otherwise 0.
- $t_i \geq 0$ : Continuous variable representing the arrival time at node  $i$ .
- $u_i \geq 0$ : Continuous variable representing the cumulative vehicle load after servicing node  $i$ .

### 2.2 Objective Function

The objective minimizes a weighted linear combination of total travel time and distance. Let  $\alpha$  and  $\beta$  be tunable scalars.

$$\min Z = \sum_{k=1}^K \sum_{i=0}^N \sum_{j=0}^N (\alpha T_{ij} + \beta D_{ij}) x_{ijk} \quad (2)$$

### 2.3 System Constraints

- **Degree/Service Constraint:** Every customer node is visited exactly once.

$$\sum_{k=1}^K \sum_{i=0, i \neq j}^N x_{ijk} = 1 \quad \forall j \in \{1, \dots, N\} \quad (3)$$

- **Flow Conservation:** A vehicle must exit any node it enters.

$$\sum_{i=0}^N x_{ihk} - \sum_{j=0}^N x_{hjk} = 0 \quad \forall h \in V, \forall k \in K \quad (4)$$

- **Capacity & Subtour Elimination (MTZ Formulation):** Cumulative load must respect maximum capacity  $C$ , and disjoint subtours are algebraically eliminated. Let  $q_j$  be the demand of node  $j$ .

$$u_j \geq u_i + q_j - C \left( 1 - \sum_{k=1}^K x_{ijk} \right) \quad \forall i, j \in \{1, \dots, N\} \quad (5)$$

$$q_i \leq u_i \leq C \quad \forall i \in \{1, \dots, N\} \quad (6)$$

- **Time-Window Enforcement:** Arrival times must respect chronological traversal and hard limits  $[e_i, l_i]$ , where  $M$  is an arbitrarily large scalar.

$$t_j \geq t_i + T_{ij} - M \left( 1 - \sum_{k=1}^K x_{ijk} \right) \quad \forall i, j \in V \quad (7)$$

$$e_i \leq t_i \leq l_i \quad \forall i \in V \quad (8)$$

### 3 Quantum-Inspired Particle Swarm Optimization (QPSO)

To solve the NP-hard MILP, the engine employs QPSO, a metaheuristic derived from quantum mechanics that provides mathematical guarantees for global convergence by breaking classical kinematic restrictions.

#### 3.1 Classical Particle Swarm Optimization (PSO)

Classical PSO operates in a  $D$ -dimensional Euclidean search space where a swarm of  $N$  particles explores an objective landscape. Each particle  $i$  is characterized by a deterministic state vector consisting of its position  $X_i^t \in \mathbb{R}^D$  and velocity  $V_i^t \in \mathbb{R}^D$  at discrete time step  $t$ . The system evolves through affine transformations driven by personal and global attractors:

$$V_i^{t+1} = \omega V_i^t + c_1 r_1 (P_i^t - X_i^t) + c_2 r_2 (G^t - X_i^t) \quad (9)$$

$$X_i^{t+1} = X_i^t + V_i^{t+1} \quad (10)$$

where  $\omega$  is the inertia weight,  $P_i^t$  is the particle's historical best position,  $G^t$  is the global swarm best,  $c_1, c_2$  are cognitive and social coefficients, and  $r_1, r_2 \sim U(0, 1)$ . The structural limitation here is the rigid reliance on a localized velocity vector, which dynamically bounds the particle's trajectory and restricts global topological exploration.

#### 3.2 The Wave Function and Delta-Potential Well

In quantum mechanics, a particle lacks a deterministic kinematic trajectory. Its state is instead represented by a wave function  $\psi(x, t)$  residing in a complex Hilbert space. The structural geometry of the particle's location is probabilistic, defined by Born's rule, where the probability density function (PDF) is  $Q(x) = |\psi(x, t)|^2$ .

If a particle is bound by a one-dimensional attractive Delta-potential well centered at an attractor  $p$ , the time-independent Schrödinger equation takes the form:

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} - \gamma \delta(x - p) \psi = E \psi \quad (11)$$

Solving this eigenvalue problem yields a normalized wave function:

$$\psi(x) = \frac{1}{\sqrt{L}} \exp\left(-\frac{|x - p|}{L}\right) \quad (12)$$

where  $L$  defines the characteristic length of the potential well. The resulting PDF is a double exponential (Laplace) distribution:

$$Q(x) = |\psi(x)|^2 = \frac{1}{L} \exp\left(-2\frac{|x - p|}{L}\right) \quad (13)$$

#### 3.3 Quantum-Behaved Particle Swarm Optimization (QPSO)

QPSO discards the classical velocity vector entirely. Particles are treated as spin-less entities moving in a quantum potential well centered on a local attractor  $P_i^t$ . By sampling the probability measure  $Q(x)$  via Monte Carlo methods, the position update collapses into a strictly stochastic operation.

The local attractor  $P_i^t$  lies on a linear segment between the personal and global bests, with  $\phi \sim U(0, 1)$ :

$$P_i^t = \phi P_i^t + (1 - \phi) G^t \quad (14)$$

The spatial extent of the potential well  $L$  is dynamically scaled by the distance between the particle and the Mean Best Position ( $mbest^t$ ), the centroid of all personal bests:

$$mbest^t = \frac{1}{N} \sum_{j=1}^N P_j^t \quad (15)$$

Evaluating the inverse cumulative distribution function of the exponential PDF yields the discrete QPSO position update. Let  $u \sim U(0, 1)$  and  $\alpha$  be the contraction-expansion coefficient scaling the topological space:

$$X_i^{t+1} = P_i^t \pm \alpha |mbest^t - X_i^t| \ln \left( \frac{1}{u} \right) \quad (16)$$

This structure allows particles to instantly sample the entire domain  $\mathbb{R}^D$  with non-zero probability, escaping classical kinematic bounds.

### 3.4 Random-Key Decoding

Because QPSO operates in continuous space  $\mathbb{R}^N$  but combinatorial routing requires discrete permutations, the continuous vector is decoded using a rank-based bijective mapping (*argsort*). Let  $\pi_i^t$  be the resulting customer visitation sequence:

$$\pi_i^t = \text{argsort}(X_i^t) \quad (17)$$

## 4 Algebraic Pathfinding Layer (A\* / Dijkstra)

To provide the QPSO algorithm with an accurate physical distance matrix, the shortest-path routing is computed at the graph layer. Let  $g \in \mathbb{R}^{|V|}$  be the accumulated cost vector, initialized as  $g_0 = 0$  and  $g_v = \infty$  for all  $v \neq 0$ .

- **Edge Relaxation (Dijkstra):**

$$g_v = \min(g_v, g_u + T_{uv}) \quad \forall (u, v) \in E \quad (18)$$

- **Heuristic Vector ( $h$ ):** For A\* search,  $h \in \mathbb{R}^{|V|}$  utilizes the  $L_2$  norm (Euclidean distance) bound by the maximum speed limit to guarantee admissibility.

$$h_v = \frac{\|v - v_{\text{target}}\|_2}{v_{\text{max}}} \quad (19)$$

- **Priority Function ( $f$ ):** The sequence of node evaluation is dictated by minimizing:

$$f_v = g_v + h_v \quad (20)$$